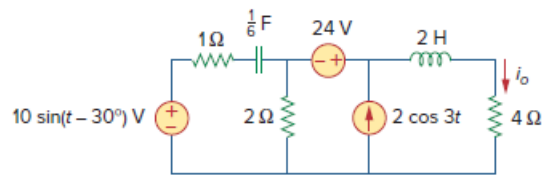


Problem

Determine i_o in the circuit of Fig. 10.92, using the superposition principle.

Figure 10.92:



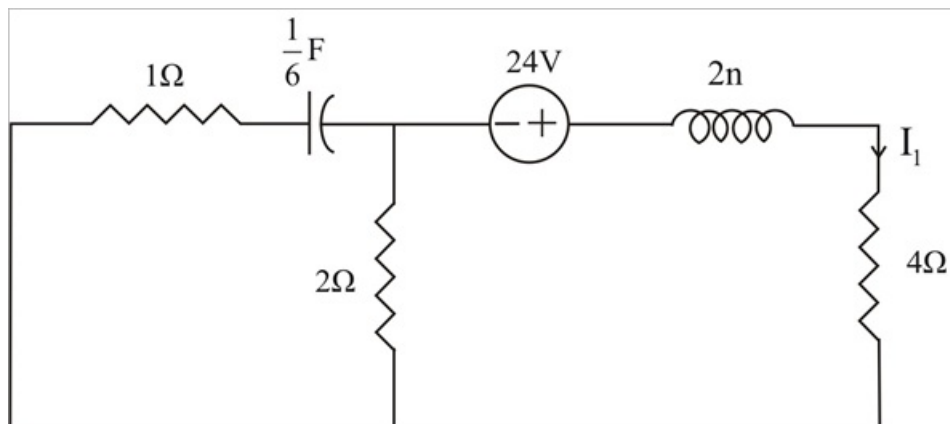
Step-by-step solution

Step 1 of 7

Let $i_o = I_1 + I_2 + I_3$ where I_1 , I_2 and I_3 are respectively due to the 24V dc source, AC voltage source and the AC current source.

Step 2 of 7

Consider 24V source only, assume the current in 4Ω resistor as I_1



Step 3 of 7

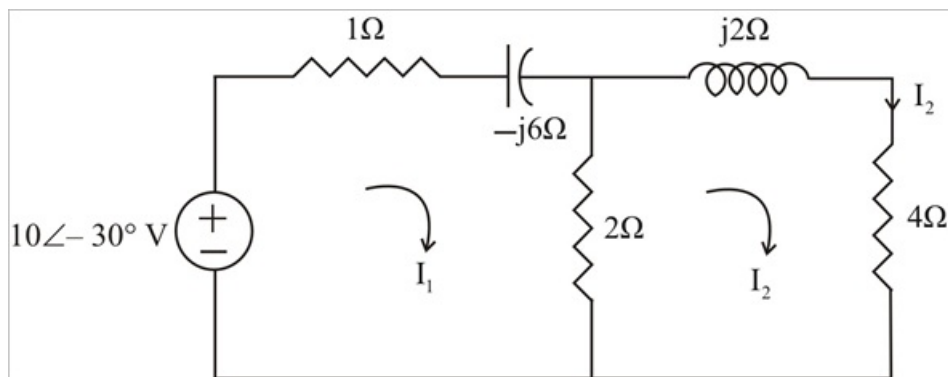
Since the capacitor is an open circuit to DC,

$$I_1 = \frac{24}{4+2}$$

$$I_1 = 4\text{A}$$

Step 4 of 7

Now consider AC voltage source alone and assume the currents as below:



[Comments \(2\)](#)



Anonymous

I believe this one has a mistake; the voltage source does not take into account that the time domain function for the voltage MUST be in cosine form in order to go directly to polar form



Anonymous

the angle should $-30 - 90 = -120$

Step 5 of 7

From the given voltage source $\omega = 1$

$$X_L = j\omega L = 2j$$

$$X_C = \frac{1}{j\omega C} = -6j$$

Mesh equations are as below:

$$-10\angle 30^\circ + (3 - j6)I_1 - 2I_2 = 0$$

$$(3 - 6j)I_1 - 2I_2 = 10\angle 30^\circ \dots\dots\dots (1)$$

$$-2I_1 + (6 + j2)I_2 = 0 \dots\dots\dots (2)$$

By solving equations (1) and (2), we can get

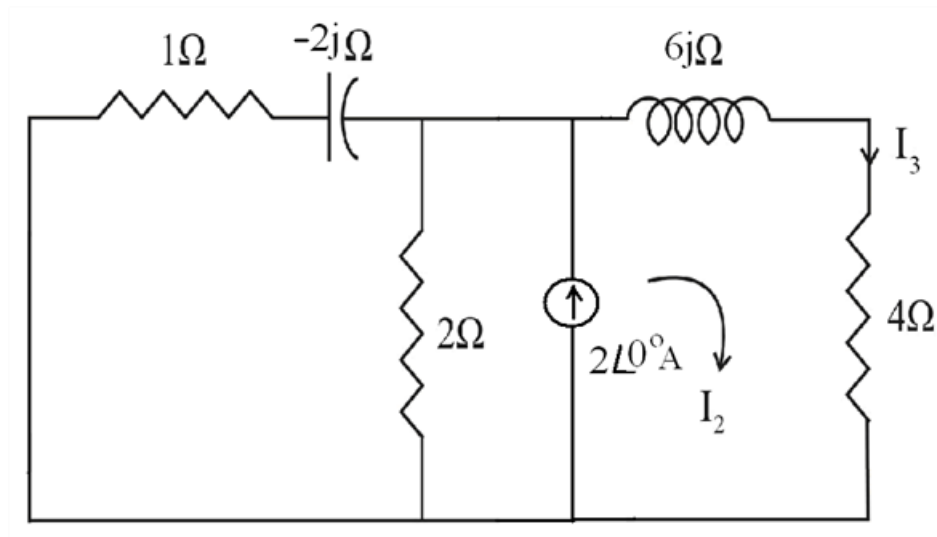
$$I_1 = 1.2636 + 0.9703j$$

$$I_2 = 0.4761 + 0.1647j$$

$$I_2 = 0.504\angle 19.1^\circ \text{A}$$

Step 6 of 7

Now consider AC current source alone



Step 7 of 7

For this ac source $\omega = 3$

$$X_L = j\omega L = 6j$$

$$X_C = \frac{1}{j\omega C} = -2j$$

$$(1 - 2j) \parallel 2 = 1.0769 - 0.6154j$$

$$I_3 = 2 \times \frac{1.0769 - 0.6154j}{1.0769 - 0.6154j + 4 + 6j}$$

$$I_3 = \frac{2.1538 - 1.2308j}{5.0769 + 5.3846j}$$

$$I_3 = 0.0786 - 0.32584j$$

$$I_3 = 0.3352 \angle -76.43^\circ \text{ A}$$

Total current in 4Ω resistor due to presence of all sources

$$i_o = 4 + 0.504 \angle 19.1^\circ + 0.3352 \angle -76.43^\circ$$

$$i_o = \boxed{4 + 0.504 \sin(t + 19.1^\circ) + 0.3352 \cos(3t - 76.43^\circ) \text{ A}}$$